

Title: Job Skill Demand Forecasting

Name: Nishant Kaushik

1. Introduction

The technology job market evolves rapidly, and skills that are in high demand today may decline tomorrow. Job seekers, recruiters, and training providers need timely information on emerging skills, regional hiring trends, and salary expectations. This project, Job Skill Demand Forecasting, demonstrates an interactive data analytics dashboard that helps users explore and forecast job skill demand across India.

2. Problem Statement

Existing information about trending skills is mostly static (blog posts, articles) and global in nature. They do not provide up-to-date, location-specific, or quantitative insights from job postings. Job seekers cannot filter by location or industry, and recruiters lack a tool to plan skill acquisition and compensation budgets. Our goal was to build a prototype dashboard that ingests job posting data, analyses demand for different skills, and visualises trends over time.

3. Objectives

- To identify skills trending or declining in demand over time.
- To visualise demand by location, industry, and work type.
- To forecast future demand for selected skills.
- To provide quick insights on salary distributions, experience requirements, top cities, top companies, and highest-paying jobs.
- To allow multi-skill comparison within one interface.

4. Methodology

A synthetic dataset of job postings for 2023–2025 (created because no real-time Indian dataset is publicly available) was used. Each record includes: Date, Skill, Job Title, Company, Location, Industry, Work Type, Salary range, and Experience required.

Steps:

1. Data Loading & Preprocessing: Used custom modules (src.data_load, src.preprocess) to load and clean the data.
2. Feature Engineering: Computed monthly skill counts, rolling trends, sector/city splits, salary stats, and experience distributions.
3. Forecasting: Used Prophet to model and forecast skill demand for the next 6 months.
4. Visualisation: Built an interactive Streamlit app with Plotly charts (line trends, heatmaps, box plots, histograms, pie charts).
5. User Interaction: Added filters for date, location, industry, work type, skill selection, and multiple skill comparison.

5. Implementation

- Front-end: Streamlit web app with tabs ("Overview", "Trends", "Forecast", "Salary & Experience", "Summary").
 - Back-end: Python 3.13 with custom src modules for loading, preprocessing, and feature computation.
 - Libraries Used: pandas, numpy, matplotlib, plotly, prophet, nltk, streamlit.
 - Dataset: synthetic_job_postings_2023_2025.csv containing ~15K records.
- Key Screens:
Overview Tab, Trends Tab, Forecast Tab, Salary & Experience Tab, Summary Tab.

6. Results & Observations

- Users can quickly identify which skills have the highest job postings in a given city or industry.
- Forecast plots provide an indicative trajectory of skill demand.
- Salary and experience stats allow better career planning.
- Multi-skill comparison enables direct benchmarking of competing skills.

7. Conclusion & Future Work

This project demonstrates a prototype of a data-driven skill demand forecasting engine. Even though synthetic data was used, the same pipeline can ingest live job posting data from APIs to become a real-time system.

Planned enhancements include:

- Integration of live job feeds from LinkedIn, Indeed, Naukri and other APIs to make the dashboard real-time.
- Recommendation of training institutes / online courses (Coursera, Udemy, local colleges) for trending skills in the user's selected location so job seekers know where to upskill.
- Automated job alerts when demand for a chosen skill rises in a city/industry, delivered via email or dashboard notifications.
- Skill gap analysis from an uploaded CV or LinkedIn profile to highlight missing competencies and recommend learning paths.
- Salary trend forecasting in addition to demand forecasting, so users can anticipate compensation changes.
- Regional comparison dashboards for comparing two cities or two industries side by side.
- Career path visualisation to show common next roles or complementary skills based on market data.

These features will turn the prototype into a comprehensive career intelligence platform rather than just a static analytics dashboard.

8. References / Tools

- Python 3.13
- Streamlit
- Plotly
- Prophet
- Synthetic Job Postings Dataset (created for demonstration)